12a Parallel Computing 50Q

AICTE Assistant Professor Screening Test - Computer Science

Topic 12

Syllabus: Parallel Virtual Machine (PVM) and Message Passing Interface (MPI) libraries and calls, Advanced architectures, Today's fastest computers.

Q1. PVM stands for:

A) Parall

Q2. MPI stands for:

A) Messa

Q3. Which of the following is a distributed-memory parallel programming standard?

A) OpenM

Q4. In Flynn's taxonomy, SISD refers to:

A) Single

Q5. GPUs are best classified as:

A) SISD

Q6. Which MPI routine must be called first in every MPI program?

A) MPI_C

Q7. Which MPI routine must be called last to cleanly exit?

A) MPI_In

Q8. In MPI, MPI_Comm_rank returns:

A) Total n

Q9. In MPI, MPI_Comm_size returns:

A) Rank o

Q10. Default communicator that includes all processes is:

A) MPI_C

Q11. Which MPI call sends a message (blocking)?

A) MPI_R

Q12. Which MPI call receives a message (blocking)?

A) MPI_S

Q13. Non-blocking send in MPI is:

A) MPI_S

Q14. MPI_Bcast is used for:

A) Point-t

Q15. MPI_Reduce is used to:

A) Reduce

Q16. MPI_Barrier does:

A) Encryp

Q17. MPI_Scatter:

A) Gather

Q18. MPI_Gather:

A) Collect

Q19. PVM was developed primarily at:

A) Micros

Q20. Which is TRUE for PVM vs MPI?

A) PVM s

Q21. In PVM, the daemon that must run on each host is called:

A) pvmd

Q22. PVM function pvm_spawn is used to:

A) Termin

Q23. PVM function pvm_send / pvm_recv are for:

A) File I/O

Q24. Amdahl's Law gives:

A) Maximu

Q25. If 90% of a program is parallelizable, with infinite processors, maximum speedup per Amdahl is:

A) 5x

Q26. Gustafson's Law (scaled speedup) argues:

A) Speedu

Q27. Speedup S(p) is defined as:

A) T_para

Q28. Efficiency E(p) is defined as:

A) S(p) / p

Q29. Which is an example of shared-memory API?

A) MPI

Q30. UMA stands for:

A) Uniform

Q31. NUMA stands for:

A) Non-Un

Q32. In a cluster, nodes are connected via:

A) High-sp

Q33. Beowulf clusters are built using:

A) Off-the-

Q34. Flynn's MIMD includes:

A) Multico

Q35. SIMD machines include:

A) Vector

Q36. Which topology has diameter log2(N) for N nodes?

A) Linear

Q37. Deadlock in message passing can occur when:

A) Two pr

Q38. MPI_TAG is used to:

A) Identify

Q39. MPI_Status object after MPI_Recv contains:

A) Source

Q40. Collective communications in MPI require:

A) All proc

Q41. Which MPI operation finds maximum value across processes?

A) MPI_R

Q42. Granularity in parallel computing refers to:

A) Ratio o

Q43. Data parallelism means:

A) Same

Q44. Task parallelism means:

A) Differen

Q45. Advanced architecture: Multi-core is:

A) Multiple

Q46. GPGPU computing commonly uses:

A) CUDA,

Q47. Today's fastest supercomputers (exascale era, e.g., Frontier, Aurora) achieve:

A) ExaFL

Q48. TOP500 list ranks supercomputers based on:

A) LINPA

Q49. Which interconnect is most common in TOP supercomputers for low latency?

A) InfiniBa

Q50. PVM supports:

A) Hetero

ANSWER KEY (12a Parallel Computing)

1-B, 2-A, 3-C, 4-A, 5-B, 6-B, 7-B, 8-B, 9-C, 10-C,

11-B, 12-E